Isabelle Kwan

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Education

Simon Fraser University — *Burnaby, B.C.*
Bachelor of Science in Computing Science (GPA 3.96/4.33) — *Expected Graduation: April 2027*
Concentration: Visual and Interactive Computing
Dean's and President's Honour Roll

Publications

C3 **Kwan, I.**, Chen, J.Z., Brehmer, M. (2026) *The Choreography of Augmented Reality Timelines: Studying the Relative Position, Chronology, Situatedness of Event Sequences. Graphics Interface 2026* https://doi.org/10.1145/3831423.3831457

C2 Liu, H., **Kwan, I.**, Okuma, T., Loverock, J., Vincent, N., & Chilana, P. (2026). *Understanding AI Literacy for Creatives: Tensions Between Structured Tutorials and Self-Experimentation. Creativity & Cognition 2026.* https://doi.org/10.1145/3803784.3807570

C1 Chaudhury, R., Huffman, C., **Kwan, I.**, Deol, G. S., Dhillon, S., & Chilana, P. K. (2025). *MILESTONES: The Design and Field Evaluation of a Semi-Automated Tool for Promoting Self-Directed Learning Among Online Learners. CHI 2025.* https://doi.org/10.1145/3706598.3714295

Experience

Research Assistant — *Sep 2026 – Present*
SFU PADComp | Python, VegaLite, DuckDB, Mosaic — *Burnaby. BC.*

- Supervisor: Dr. Nicholas Vincent
- Designing interactive visualizations for a large-scale simulation study investigating collective action and data migration behavior across competing platforms.
- Designing and conducting user studies for PaperMuncher, a LLM API-based tool to convert papers between different disciplinary forms.

Research Intern — *Sep 2025 – Dec 2025*
UW HCI Lab · Undergraduate Research Fellowship | C#, ARCore, Unity — *Waterloo. ON.*

- Supervisor: Dr. Matthew Brehmer
- Developed a tablet-based AR application in **Unity** for 3D spatial interaction and data exploration.
- Conducted user studies and analyzed user interactions with AR tools, 3D timelines, and varying datasets.

Software Engineering Intern — *Sep 2024 – Apr 2025*
Excelar Technologies | Python, Node.js, PHP, PostgreSQL, MySQL, Redis, Docker — *Burnaby. B.C.*

- Designed scalable APIs, improving access speed by **25%** and supporting seamless user experiences.
- Architected and implemented robust data models and backend systems, optimizing database schemas, reducing redundancy, streamlining data workflows, and enhancing query performance by **20%**.
- Refactored ETL pipelines, increasing data processing efficiency by **15%** and enhancing maintainability.

Research Intern — *May 2024 – Sep 2024*
SFU Interactive Experiences Lab · NSERC USRA | Javascript, VEGA-Lite, React.js — *Burnaby. B.C.*

- Supervisor: Dr. Parmit Chilana
- Created scripts to convert user's "study-habits" into interactive visualizations using **VEGA-Lite**.
- Reviewed and sourced relevant literate for several projects in the research lab.
- Managed 4 projects by facilitating meetings, creating agendas, and assisting in conducting user studies.

Awards and Recognitions

Computing Research Association Outstanding Undergraduate Researchers Nominee *2026*

- One of four nominees from Simon Fraser University, award currently under review.

EA Dick Asher Award in the Faculty of Applied Sciences — $13,600 *2026*

- Recognized for top academic achievements and work experience.

Improving Award for Women In Computing Science — $2,500 *2026*

- Recognized for excellent academic standing and outstanding community service.

Grace Hopper Conference Scholar — $2,400 *2025*

- One of five, recognized for excellent academic standing and outstanding community service.

CIBC Undergraduate Award for Women in Faculty of Applied Sciences — $5,000 *2025*

- One of three, recognized for engagement in technology-focused work and impact in technology-focused communities, activities, and student organizations.

Computing Science Undergraduate Student Society Award — $3,000 *2023*

- One of three, Recognized for exceptional contributions to SFU through participation and volunteering with the Computing Science Student Society, Women in Computing Science, and Peer Tutoring.

SFU Major Entrance Scholarship with Distinction — $27,000 *2022*

- Awarded for being within the top 1% of applicants in academics, community service, and achievements

BC Excellence Award (Provincial) — $5,000 *2022*

- One of 55 students in British Columbia who demonstrates service and leadership, both at school and in their communities, and who show aptitude for and commitment to their chosen career paths.

Governor General's Bronze Academic Medal *2022*

- Recognized for achieving the highest grade point average upon graduating from a Canadian secondary school institution.

Volunteer Experience

Director of Graduate Events | SFU Women in Computing Science *May 2026 – Present*
Senator | Simon Fraser University *May 2025 – Present*
Co-Founder and Host | CAN-CWIC West Conference *June 2024 – Present*
President | SFU Computing Science Student Society *May 2024 – Apr 2025*